Does National Individualism Predict AI Sentiment?

Testing the Cultural-Dimensions Explanation for the Asia-West AI Optimism Gap

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Abstract

Global surveys of public attitudes toward AI consistently find a striking regional pattern: countries such as Indonesia and Thailand report overwhelming optimism about AI, while countries such as Canada and the United States are far more skeptical. Cross-cultural psychology offers a candidate explanation: individualistic cultures tend to frame new technology as a potential threat to personal autonomy, while collectivist cultures more often frame it as a tool that helps people function within their social environment. This analysis tests that explanation directly and quantitatively, correlating Hofstede's Individualism Index against Ipsos's 2025 AI Monitor survey. While checking the underlying cultural-dimension data, a substantive methodological issue was uncovered: the current official Hofstede Insights source uses a different, non-comparable methodology for this specific dimension than the classic scores used

in most published research – a distinction Hofstede's own successor organisation explicitly warns about. The primary analysis therefore uses only the 24 countries whose classic IDV score could be confirmed against a single consistent source, finding a strong, statistically robust negative correlation ($r=-0.83$, $p=5.2\times10^{-7}$): individualism alone explains approximately 69% of the cross-country variation in AI sentiment among these 24 countries.

1 Introduction and Motivation

Public commentary on the global divide in AI sentiment tends to reach for economic explanations (wealthier countries have more to lose; poorer countries see AI as a path to catching up) or narrower cultural anecdotes (Japan's friendly-robot pop culture versus Hollywood's dystopian AI narratives). This analysis tests a more specific, established psychological construct directly: Hofstede's Individualism Index, a widely used measure of whether a society emphasizes individual autonomy or group interdependence. The hypothesis, drawn from published cross-cultural psychology research, is that individualistic societies frame AI primarily as a threat to personal autonomy and privacy, while collectivist societies more often frame it as an extension of the self that helps one function within a group – a positive frame in a collectivist value system.

2 Research Question

Does Hofstede's Individualism Index (IDV) correlate with the percentage of a country's population who see AI as more beneficial than harmful?

3 Data and Limitations

AI sentiment. Ipsos AI Monitor 2025, a 30-country Global Advisor survey of 23,216 adults, fielded March 21 - April 4, 2025. The specific item used is "Products and services using artificial intelligence have more benefits than drawbacks."

Cultural dimension. Hofstede's Individualism Index (IDV), a 0-100 scale where higher values indicate greater individualism.

Known limitations.

- **This Ipsos wave does not include China.** The widely cited "China 83%" AI-optimism figure comes from a different Ipsos wave (cited in Stanford's AI Index 2025), not the 2025 30-country wave used here.
- **Two incompatible measures are both called "Hofstede Individualism."** This is the most important limitation of this analysis, and it materially changed the reported result. While sourcing IDV scores for all 30 Ipsos countries, six (Thailand, Singapore, Ireland, Sweden, Hungary, Spain) could not be confirmed against the same single, internally consistent peer-reviewed source used for the other 24. Attempting to resolve this by checking the current official Hofstede Insights website revealed a deeper problem: that source switched several years ago to a different underlying methodology for this specific dimension, based

on a 2022 study by Minkov and Kaasa. Hofstede's own successor organisation (geerthofstede.com) states this directly: the newer measure produces "counter-intuitive results" (for example, rating Japanese culture as more individualistic than American culture, which contradicts the classic scores used throughout most published research), and that "Minkov-Kaasa-Individualism is actually a different concept than Hofstede's Individualism, with different data behind it." Using the current official website to fill the six missing scores would therefore not have resolved the data-quality gap – it would have silently combined two non-comparable constructs under one column name. For this reason, the **primary analysis in this report uses only the 24 countries with a confirmed classic-IDV score**, and the six affected countries are excluded from the headline result rather than patched with data of uncertain comparability.
- **Sample representativeness.** Ipsos's own methodology discloses that several countries in the survey (including Brazil, Indonesia, Malaysia, Thailand, and Turkiye) have samples skewed toward more urban, educated, or affluent respondents than their general population.
- **Correlational, not causal.** A strong correlation between individualism and AI sentiment does not establish that cultural values cause the sentiment gap. Both could be driven by a shared third factor – for example, economic growth expectations, government AI strategy and messaging, or media environment – that this analysis does not measure directly.

4 Methodology

1. **Clean and merge** (`analysis.R`, Part 1): the two datasets were joined on country name. Each row was flagged according to whether its IDV score came from the single consistent classic-IDV source. The primary analysis dataset was restricted to the 24 countries passing this check.
2. **Test** (Part 2): Pearson and Spearman correlations were computed between IDV and AI sentiment on the primary (n=24) dataset, followed by a simple linear regression. The result of including the six excluded countries using their unconfirmed-source values is also printed to `stats_results.txt`, explicitly labeled as not a reported finding, to document that it does not change the qualitative conclusion.

5 Results

Primary analysis: n = 24 countries with a confirmed classic Hofstede IDV score.

Table 1: Summary of statistical tests, primary analysis (n=24).

Test	Result
Pearson correlation	r = -0.83, $p = 5.2 \times 10^{-7}$
Spearman correlation	rho = -0.82, $p = 9.3 \times 10^{-7}$
Linear model	$R^2 = 0.69$, slope = -0.39

Test	Result

The relationship is visualized in Figure 1: a strong, clearly linear, negative relationship, with the most collectivist countries in the sample (Indonesia, Malaysia, Peru, South Korea) clustered at the high-sentiment end, and the most individualistic countries (the United States, Australia, the United Kingdom, Canada) clustered at the low-sentiment end.

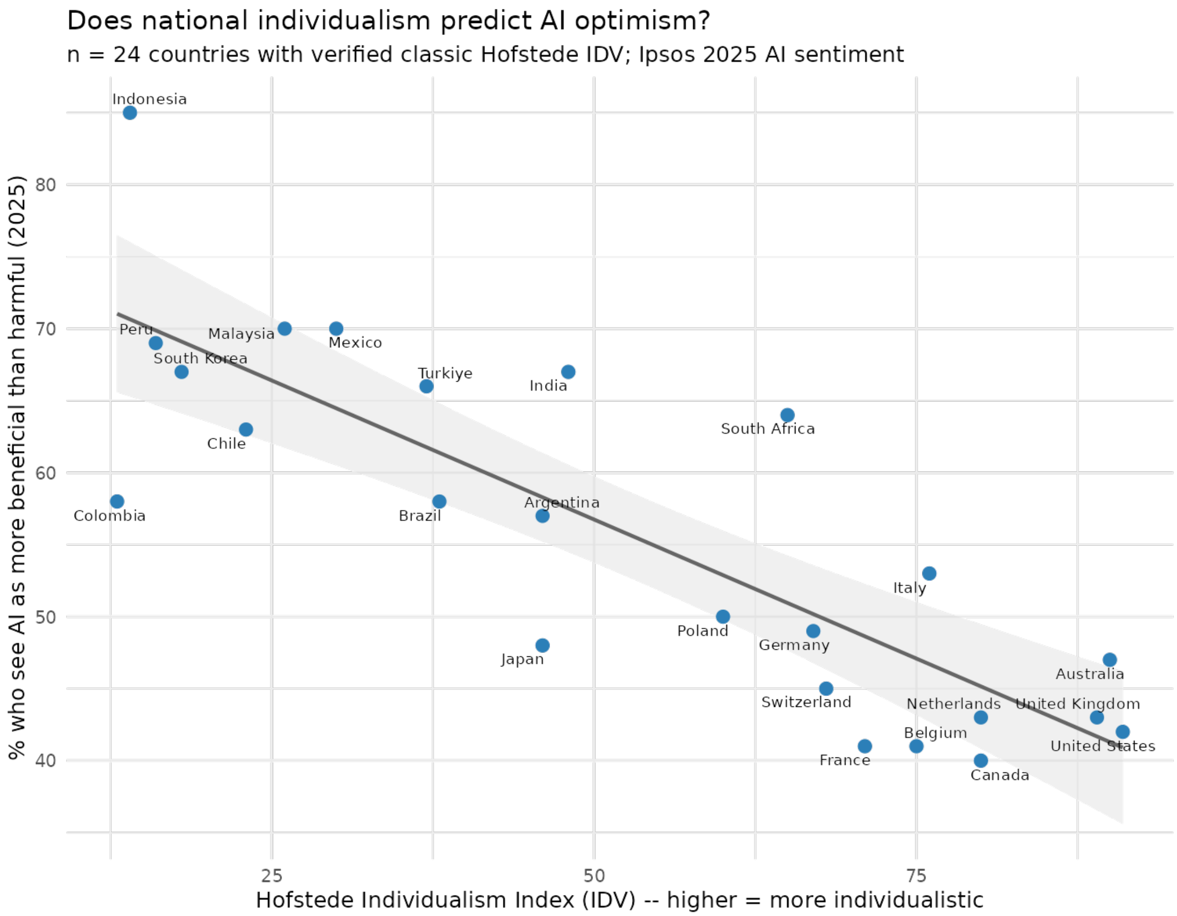

Figure 1: Hofstede Individualism Index vs. Ipsos 2025 AI sentiment, 24 countries with a confirmed classic IDV score.

Individualism alone explains approximately 69% of the variance in AI sentiment across these 24 countries (R^2=0.69). For transparency, `stats_results.txt` also reports what the correlation looks like if the six excluded countries are added back in using their unconfirmed-source values (r=-0.85, n=30) – included only to document that the primary conclusion is not sensitive to their exclusion, not as an alternative valid estimate, since doing so would mix potentially incompatible IDV measurement approaches.

6 Discussion

The strength of this correlation (R^2=0.69) is worth treating with appropriate caution precisely because it is so strong. Cross-national social-attitude research rarely finds a single cultural variable explaining more than two-thirds of the variance in a complex, multiply-determined attitude like technology sentiment. Three considerations are worth holding alongside the headline number.

First, individualism is very likely correlated with other national characteristics that plausibly also shape AI sentiment – GDP per capita, media environment, government AI strategy and public messaging, and existing familiarity with AI tools through daily life. This analysis tests a bivariate relationship; it does not control for these other factors.

Second, the data-quality investigation required to build this dataset properly turned out to be a finding in its own right. Discovering that Hofstede's own successor organisation has publicly disavowed comparability between the classic IDV measure and the current official website's Individualism scores is a genuinely useful methodological note for anyone doing quantitative cross-cultural research: a citation to "Hofstede's Individualism Index" is no longer sufficient on its own to specify which measure is meant, and mixing the two without checking would produce a dataset that looks complete but combines non-comparable numbers under a single column name.

Third, the exclusion of China from this particular Ipsos wave is a meaningful gap. China is frequently cited as having the single highest AI optimism (83% in a separate Ipsos wave) and, under the classic Hofstede framework, one of the most collectivist cultures measured (IDV=20). Its absence from the 30-country sample used here does not undermine the finding, since the pattern holds clearly across the 24 countries analyzed, but a verified China data point would have strengthened the sample's coverage at exactly the extreme end of both variables.

7 Limitations and Future Work

1. **Multivariate modeling.** A regression including GDP per capita, government AI strategy indicators, and media environment measures alongside individualism would test whether the cultural-dimension effect survives controlling for these plausible confounds.
2. **Resolving the six excluded countries properly.** Locating classic-IDV-specific values (not the newer Minkov-Kaasa measure) for Thailand, Singapore, Ireland, Sweden, Hungary, and Spain from an academic source using the original IBM-survey methodology would allow them to be added back to the primary analysis with confidence.
3. **Including China.** Locating a wave of the Ipsos AI Monitor (or a comparable survey) that includes both China and the other countries in a single consistent administration would allow the extreme end of both variables to be tested directly within one dataset.
4. **Longitudinal tracking.** Since Ipsos fields this survey annually, tracking whether the individualism-sentiment relationship strengthens, weakens, or

holds steady as AI becomes more familiar globally would be a natural extension.

8 Conclusion

Hofstede's classic Individualism Index is a strong, statistically robust predictor of national AI sentiment across the 24 countries in this analysis with a confirmed score, explaining approximately 69% of the cross-country variance. This is consistent with – though not sufficient on its own to prove – the cross-cultural psychology explanation that individualistic societies frame AI primarily as a threat to personal autonomy, while collectivist societies more often frame it as a tool that helps people function within their social environment. Building this dataset also surfaced a genuine methodological finding: two incompatible measures currently share the name "Hofstede Individualism," and treating them as interchangeable would silently corrupt any analysis that mixes them.

References

Ipsos. (2025). *The Ipsos AI Monitor 2025: A 30-country Ipsos Global Advisor Survey*. Retrieved from ipsos.com

Hofstede, G. (1997). *Cultures and Organizations: Software of the Mind*. McGraw-Hill.

Minkov, M., & Kaasa, A. (2022). A test of Hofstede's model of culture following his own approach. *Journal of International Management*, 28(4).